

Number Fundamentals – Homework Practice Sheet

1) Check Palindrome Number

Problem Statement

Given a number , check whether the number is a palindrome or not.

Sample Input 1

121

Expected Output 1

Palindrome

Sample Input 2

123

Expected Output 2

Not Palindrome

2) Find Largest Digit in a Number

Problem Statement

Given a number , return the largest digit present in the number.

Sample Input 1

58391

Expected Output 1

9

Sample Input 2

2047

Expected Output 2

7

3) Find Smallest Digit in a Number

Problem Statement

Given a number `n`, return the smallest digit present in the number.

Sample Input 1

58391

Expected Output 1

1

Sample Input 2

2047

Expected Output 2

0

4) Count Even Digits in a Number

Problem Statement

Given a number `n`, count how many even digits are present in the number.

Sample Input 1

24891

Expected Output 1

3

Sample Input 2

13579

Expected Output 2

0

5) Count Odd Digits in a Number

Problem Statement

Given a number , count how many odd digits are present in the number.

Sample Input 1

24891

Expected Output 1

2

Sample Input 2

24680

Expected Output 2

0

6) Check Armstrong Number

Problem Statement

Given a number , check whether it is an Armstrong number or not.

Sample Input 1

153

Expected Output 1

Armstrong

Sample Input 2

123

Expected Output 2

Not Armstrong

7) Find GCD (Greatest Common Divisor)

Problem Statement

Given two numbers `a` and `b`, find their GCD.

Sample Input 1

12 18

Expected Output 1

6

Sample Input 2

20 50

Expected Output 2

10

8) Find LCM (Least Common Multiple)

Problem Statement

Given two numbers and , find their LCM.

Sample Input 1

12 18

Expected Output 1

36

Sample Input 2

4 6

Expected Output 2

12

9) Check Perfect Number

Problem Statement

Given a number `n`, check whether it is a perfect number or not. A **perfect number** is a **positive integer** that is **exactly equal to the sum of its proper positive divisors** (excluding the number itself).

Sample Input 1

6

Expected Output 1

Perfect Number

Sample Input 2

10

Expected Output 2

Not Perfect Number

10) Find Power of a Number

Problem Statement

Given two numbers `a` and `b`, calculate `a^b`.

Sample Input 1

2 5

Expected Output 1

32

Sample Input 2

3 4

Expected Output 2

81

11) Count Zeros in a Number

Problem Statement

Given a number `n`, count how many zeros are present in the number.

Sample Input 1

102030

Expected Output 1

3

Sample Input 2

9999

Expected Output 2

0

12) Check Spy Number

Problem Statement

Given a number , check whether the sum of digits and product of digits are equal.

Sample Input 1

123

Expected Output 1

Spy Number

Sample Input 2

145

Expected Output 2

Not Spy Number

13) Find Sum of Factors

Problem Statement

Given a number `n`, return the sum of all factors of the number.

Sample Input 1

6

Expected Output 1

12

Sample Input 2

10

Expected Output 2

18

14) Count Prime Digits in a Number

Problem Statement

Given a number `n`, count how many prime digits are present in the number.

Prime digits are: 2, 3, 5, 7.

Sample Input 1

23789

Expected Output 1

4

Sample Input 2

4680

Expected Output 2

0

15) Find First and Last Digit Sum

Problem Statement

Given a number `n`, return the sum of the first digit and last digit.

Sample Input 1

12345

Expected Output 1

6

Sample Input 2

9081

Expected Output 2

10